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AI-Powered Secure Decentralized File Sharing System Using Blockchain and IPFS

A MAIN PROJECT PHASE 1 REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE

REQUIREMENTS FOR THE AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY in

Computer Science and Engineering of

APJ Abdul Kalam Technological University by

Yadukrishnan N K (VAS23CS197)
Mohamed Shahal (VAS23CS133)
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Certificate

This is to certify that the Main Project Phase 1 Report titled **"AI-Powered Secure DecentralizedFileSharingSystemUsingBlockchainandIPFS"**isabonafide recordoftheworkcarriedoutby**YadukrishnanNK(VAS23CS197),Mohamed Shahal(VAS23CS133),MohammedNabeelAA(VAS23CS134),Muhammed Hafis(VAS23CS138)**ofVidyaAcademyofScience&Technology,Thalakkottukara, Thrissur-680501inpartialfulfillmentoftherequirementsfortheawardof**Degree ofBachelorofTechnology**in**ComputerScienceandEngineering**of**APJAbdul KalamTechnologicalUniversity**,duringtheacademicyear2023-2027.

6

TheMain Project report has been approved as it satisfies the academic requirements in the respect of Main Project work prescribed for the said degree

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Thrissur
October 2026

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October 2026 Thrissur - 680 501.

Computer Science and Engineering Engg. i VAST, Thalakotukara

Abstract

In the modern digital environment, files are continuously shared across users, devices, teamsandnetworks,whichmakesfilesharingahigh-valuetargetforattackers.Con-ventionalcentralizedplatformsrelyonasingleserverandasingletrustboundary,soa compromised server or account can expose many files at once, while basic upload work-flows may accept malicious or sensitive content without adequate screening.Application logs on such platforms can also be edited or incomplete, which weakens incident inves-tigationandregulatorycompliance.Toaddressthesechallenges,thisprojectpresents AegisCyberShield,anAI-poweredsecuredecentralizedfilesharingsystemthatcom-bines Zero Trust access control,AI-assisted threat screening,hybrid encryption,decen-tralized storage and blockchain-based integrity records in a single file-sharing workflow. Everyrequestisverifiedthroughasix-stageZeroTrustgatewaycoveringtokenverifi-cation, IP allow-lists, session limits, role and attribute policies, behavioural scoring and AIriskassessment.Uploadedfilesareinspectedbyan AIsecurityengineformalware indicators,highentropy,sensitivepersonaldataandhiddencontent,andarethenen-crypted with AES-256-GCM, with the per-file key wrapped using RSA-4096-OAEP. The encryptedcontentispinnedtoIPFSandidentifiedbyaContentIdentifier(CID),while the SHA-256 hash,CID and digital signature are registered in a Solidity smart-contract registry. A provenance record for each operation, a security operations dashboard and au-tomated compliance scoring against GDPR, ISO/IEC 27001 and NIST CSF 2.0 support traceability and investigation.The main contribution of the project is an integrated and testableupload-to-retrievalworkflowratherthantheinventionoftheunderlyingtech-nologies.ThisPhase1reportpresentstherequirements,literaturesurvey,architecture,

data flow, module design, current implementation status and testing plan of the system.

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Vision of the Institution

Progress through education.

Mission of the Institution

To seek, strive for and scale greater heights of quality education.

M1: Seek to foster in every student a profound and practical understanding of knowledge, encouraging critical inquiry and intellectual curiosity.

M2:Strivetoensureeverystudentdevelopsholistically,cultivatingtheskills,ethical character,andresiliencenecessarytothriveindiverseandchallengingfutureenvironments.

M3:Scalethecommitmenttoservicebyinspiringstudentstoapplytheirknowledge and skills to actively address community needs, promoting ethical leadership and social responsibility.

Vision of the Department

Progressthroughqualityeducationandbecomeacenterofexcellenceincomputing knowledge.

Mission of the Department

M1:To seek and strive for advancements in technological domains.

M2:To create socially responsible engineers in the field of computer science.

M3:To bridge the gap between science and mankind through meaningful innovation and service.

Program Educational Objectives (PEOs)

PEO-1:Graduates shall practice computer science and engineering in IT industry, public sectorundertaking,asanentrepreneurorasaresearcherpursuingasuccessfulprofessional career.

PEO-2:Graduatesshalldemonstratecreativity,developinnovativeideasandworkin teams to accomplish a common goal.

PEO-3:Graduatesshalldemonstrateprojectmanagementskills,lifelonglearningand social awareness upholding ethical values.

Program Outcomes (POs)

PO1	**EngineeringKnowledge:**Applyknowledgeofmathematics,natural science, computing, engineering fundamentals and an engineering spe-cialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	**Problem Analysis:**Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclu-sions with consideration for sustainable
PO3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop sys-tems/components/processes to meet identified needs with consideration
PO4	**ConductInvestigationsofComplexProblems:**Conductinvestiga-tions using research-based knowledge including design of experiments, modelling, analysis and interpretation of
PO5	**EngineeringToolUsage:**Create,selectandapplyappropriatetech-niquesandmodernengineeringtoolsrecognizingtheirlimitations. (WK2 and WK6)
PO6	**TheEngineerandTheWorld:**Analyzesocietalandenvironmental impactswithreferencetosustainability,economy,health,safety,and culture.(WK1, WK5, WK7)
PO7	**Ethics:**Applyethicalprinciples,professionalethics,andadhereto laws.(WK9)
PO8	**Individual and Collaborative Team Work:** Function effectively as an individual and in teams.
PO9	**Communication:** Communicate effectively through reports, documen-tation, and presentations considering diverse audiences.

PO10	**ProjectManagementandFinance:**Applymanagementprinciples and economic decision-making in projects.
PO11	**Life-Long Learning:** Recognize the need for independent and lifelong learning.(WK8)

Knowledge and Attitude Profile (WK)	
WK1	A systematic, theory-based understanding of the natural sciences appli-cable to the discipline and awareness of relevant social
WK2	Conceptually-basedmathematics,numericalanalysis,dataanalysis, statisticsandformalaspectsofcomputerandinformationsciencetosupport detailed analysis and modelling applicable to the
WK3	Asystematic,theory-basedformulationofengineeringfundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks andbodiesofknowledgefortheacceptedpracticeareasintheengi-
WK5	Knowledge,includingefficientresourceuse,environmentalimpacts, whole-lifecost,re-useofresources,netzerocarbon,andsimilarcon-cepts,thatsupportsengineeringdesignandoperationsinapractice area.
WK6	Knowledgeofengineeringpractice(technology)inthepracticeareas in the engineering discipline.
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional respon-sibility of an engineer to public safety and

Knowledge and Attitude Profile (WK) (Continued)	
WK8	Engagement with selected knowledge in the current research literature ofthediscipline,awarenessofthepowerofcriticalthinkingandcre-
WK9	Ethics,inclusivebehaviorandconduct.Knowledgeofprofessional ethics, responsibilities, and norms of engineering practice.Awareness oftheneedfordiversitybyreasonofethnicity,gender,age,physical abilityetc.withmutualunderstandingandrespect,andofinclusive

Program Specific Outcomes (PSOs)

PSO-1	Theabilitytoapplytheknowledgeofcomputationalstructuresand algorithmsforsolvingcomplexengineeringproblemsinmultidisci- plinary areas.
PSO-2	Theabilitytodesign,developandevaluatecomputer-basedsystems using advanced tools and technologies to meet real-world

Sustainable Development Goals (SDGs)

SDG 1	**No Poverty:**End poverty in all its forms everywhere.
SDG 2	**Zero Hunger:**End hunger, achieve food security and improved nutri-tion and promote sustainable agriculture.
SDG 3	**Good Health and Well-being:** Ensure healthy lives and promote well-being for all at all ages.
SDG 4	**Quality Education:**Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.
SDG 5	**GenderEquality:**Achievegenderequalityandempowerallwomen and girls.
SDG 6	**CleanWaterandSanitation:**Ensureavailabilityandsustainable management of water and sanitation for all.
SDG 7	**Affordable and Clean Energy:**Ensure access to affordable, reliable, sustainable and modern energy for all.
SDG 8	**DecentWorkandEconomicGrowth:**Promotesustained,inclusive and sustainable economic growth, full and productive employment and decent work for all.
SDG 9	**Industry,InnovationandInfrastructure:**Buildresilientinfrastruc -ture,promote inclusive and sustainable industrialization and foster in-novation.
SDG 10	**Reduced Inequalities:** Reduce inequality within and among countries
SDG 11	**Sustainable Cities and Communities:**Make cities and human settle-ments inclusive, safe, resilient and sustainable.
SDG 12	**Responsible Consumption and Production:**Ensure sustainable con-sumption and production patterns.

SDG 13	**Climate Action:**Take urgent action to combat climate change and its impacts.
SDG 14	**Life Below Water:**Conserve and sustainably use the oceans, seas and marine resources for sustainable development.
SDG 15	**LifeonLand:**Protect,restoreandpromotesustainableuseofterres-trialecosystems,sustainablymanageforests,combatdesertification, and halt and reverse land degradation and halt biodiversity loss.
SDG 16	**Peace, Justice and Strong Institutions:**Promote peaceful and inclu-sive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all lev-els.
SDG 17	**Partnerships for the Goals:**Strengthen the means of implementation and revitalize the global partnership for

SDGs addressed with Justification

Project Title	SDG addressed	Justification
Aegis Cyber Shiel d	SDG9:Industry, Innovation and Infrastructure	The project integrates Zero Trust access control, AI-basedthreatscreening,hybridencryption,IPFSand blockchainintoonesecuredigitalinfrastructure.It contributes to resilient and innovative
	SDG16:Peace, Justice and Strong Institu-tions	Tamper-evident hash registration, per-provenancerecordsandachain-of-custodytrail supportaccountability,forensicinvestigationand transparentinstitutions. Encryptionbeforestorage

Project Title	SDG addressed	Justification
	SDG8: Decent WorkandEco-nomic Growth	Reducing the exposure of shared business files lowers theriskandcostofdatabreaches.Automatedcom-pliancescoringreducesmanualauditeffortandlets

Project SDGs: Relevance and Justification

4

SDG 3: Good Health and Well-being

Ensure healthy lives and promote well-being for all at all ages.

The current project does not implement a healthcare feature or demonstrate a direct health outcome. Its security mechanisms could be relevant to protecting sensitive files in a future healthcare deployment, but this should be treated as a possible application rather than a demonstrated contribution.

4

SDG 4: Quality Education

Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

The platform could be used to share academic documents and learning resources with access controls. The report does not describe a dedicated education module, so this is a potential use of the system rather than a measured educational impact.

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SDG 9: Industry, Innovation and Infrastructure

Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.

This is the strongest alignment. The project combines blockchain-based integrity records with IPFS-based decentralized storage and security controls to explore a more resilient and auditable approach to digital file sharing.

POs & PSOs addressed with Justification

Project Title	POs & PSOs addressed	Justification
Aegis Cyber Shield	PO1–Engineering Knowledge	Applies cryptography (AES, RSA, SHA-256, ECDSA),distributedsystems,blockchainandmachine learning fundamentals to the problem of secure file sharing
	PO2–Problem Analysis	Identifies weaknesses of centralized file sharing such assinglepointsoffailure,mutablelogsandunscreeneduploads,andanalyzesthemagainstZero Trust and compliance requirements.
	PO3 – Design/Development of Solutions	Designsafive-layerarchitecturewithseparatetrust boundaries for file bytes, integrity references and operational metadata.
	PO4–Conduct Investigations of Complex Problems	Reviews30researchpapersandstandards,defines testcasesforintegrity,accesscontrolandthreat screening, and plans measurement of results.
	PO5 – Modern Tool Usage	UsesReact,TypeScript,Express.js,FastAPI,Solidity,Hardhat,IPFS(Pinata)andMongoDB,andrecognizes their limitations such as IPFS availability and on-chain privacy.
	PO6–TheEngineerandTheWorld	Protectsconfidentialfiles, supportsGDPRand ISO/IEC 27001 aligned controls, and reduces the societal and economic impact of data breaches
	PO7 – Ethics	Keepspersonaldataandrawfilecontentoff-chain, avoidsexaggeratedclaimsaboutcertification,and records user actions transparently.

Project Title	POs & PSOs addressed	Justification
	PO8–Individual and Collaborative	Thefrontend,gateway,AIengine,smartcontracts and storage modules are shared among the four team members and integrated together.
	PO9 – Communication	Producesarchitecturediagrams,data-flowdescriptions, a weekly project diary, a report and review pre-sentations.
	PO10–Project Managementand Finance	Followsaseventeen-weekplanwithweeklymilestones,moduleintegration,testinganddocumentation checkpoints.
	PO11 – Life-long Learning	Works with fast-moving areas such as blockchain, decentralizedstorage,AIsecurityanalyticsandpost-

Project Title	POs & PSOs addressed	Justification
	PSO1	Applycomputationalstructuresandalgorithmsto solve complex problems. Useshashing,hybridencryption,entropyanalysis, risk scoring and smart-contract logic.
	PSO2	Design,developandevaluatecomputer-basedsystems using advanced technologies. Developsafull-stack,multi-servicesecurefile-sharingsystemthatintegratesAI,blockchainand IPFS.

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List of Abbreviations

AES Advanced Encryption Standard

ABAC	Attribute-Based Access Control
AI	Artificial Intelligence
API	Application Programming Interface
APT	Advanced Persistent Threat
CID	Content Identifier
DID	Decentralized Identifier
ECDSA	Elliptic Curve Digital Signature Algorithm
EVM	Ethereum Virtual Machine
GCM	Galois/Counter Mode
GDPR	General Data Protection Regulation
IPFS	InterPlanetary File System
JWT	JSON Web Token
KEM	Key Encapsulation Mechanism
MCP	Model Context Protocol
MFA	Multi-Factor Authentication
ML	Machine Learning
OAEP	Optimal Asymmetric Encryption Padding
PII	Personally Identifiable Information
RBAC	Role-Based Access Control

RSA	Rivest-Shamir-Adleman
SHA	Secure Hash Algorithm
SOC	Security Operations Centre
UEBA	User and Entity Behaviour Analytics
UI	User Interface
WSS	WebSocket Secure

Chapter 1

INTRODUCTION

1.1 General

Digital files are routinely shared between individuals, teams, devices and organizations, and the volume of sensitive information that travels through file-sharing services grows everyyear.Documents,sourcecode,financialrecords,personaldataandresearchmaterial are all exchanged through web platforms and cloud storage.This convenience has made file sharing one of the most attractive targets for attackers. Escalating cyber threats, ranging from credential theft and ransomware to insider misuse, mean that a file-sharing service must do much more than authenticate a user at login.The cost of failure is also high: the IBM Cost of a Data Breach Report 2023 places the global average cost of a data breachatUSD4.45million.Afilethatissharedmustthereforebechecked,protected and verified throughout its entire life cycle.

Most widely used platforms still follow a centralized design in which a single server, orasinglecloudaccount,storesthefilesandthedatabasethatdescribesthem.Sucha designcreatesasinglepointoffailureandasingletrustboundary.Iftheserveroran administrator account is compromised, a large number of files may be exposed together, especially when files are stored in a form that the server itself can read.The upload path ofabasicplatformmayacceptmaliciousorsensitivefileswithoutadequatescreening, and the platform usually offers the user no independent way to confirm that a downloaded

fileisidenticaltothefilethatwasuploaded.Accesscontrolisoftenlimitedtostatic roles,whichcannotrepresentcontextsuchasanexpiredshare,anunusualsession,an unfamiliar location or abnormal behaviour.Audit logs kept inside the same application

databasemaybeeditedorincomplete,whichweakensincidentinvestigation.Finally, tracking compliance with frameworks such as GDPR, ISO/IEC 27001 and NIST CSF is still largely a manual activity that is difficult to keep consistent.

Recent technologies provide building blocks that address parts of this problem.The ZeroTrustmodel,describedinNISTSP800-207,requireseveryaccessrequesttobe verifiedusingidentity,deviceandpolicyratherthannetworklocation.TheInterPlane-taryFileSystem(IPFS)providescontent-addressedstorageinwhichafileisidentified by a Content Identifier (CID) derived from its content.Blockchain provides an append-only, tamper-evident ledger that can record selected file hashes and access events for later verification.Artificialintelligenceanddata-analyticstechniquescanflagsuspiciousor sensitive content before it is shared. Individually, these technologies are well established, but each of them solves only one part of the problem.IPFS alone does not give confiden-tiality or access control, a blockchain alone does not store files, and an AI scanner alone does not protect stored data.

This project,named Aegis Cyber Shield,explores these controls together in one se-curefile-sharingprototype.EveryrequestisverifiedbyaZeroTrustgateway.Each uploadedfileisscannedbyanAIsecurityengine,encryptedusingahybridscheme, stored on IPFS, and registered on a blockchain registry through its hash and CID. Shar-ing,expiry,downloadlimitsandQR-basedlinksareenforcedoneverydownload,and each operation produces a provenance record that can be used for forensics and compli-ance reporting.The project does not claim to invent these technologies; its contribution istheintegrationofthecontrolsintoatraceableupload-to-retrievalworkflowtogether with a plan to evaluate its behaviour under defined test cases.As an academic prototype, it does not claim enterprise-scale security or formal compliance certification.

1.2 Objective of the Work

Theprimaryobjectiveofthisprojectistodesignanddevelopasecure,decentralized andverifiablefile-sharingsysteminwhichaccessvalidation,threatscreening,encryp-tion, decentralized storage and tamper-evident activity records operate as one

connected workflow rather than as unrelated features.The system aims to reduce the exposure that results from a single central server, to provide users with an independent integrity check,

andtogivesecurityteamsatraceablerecordofwhathappenedtoeachfile.Thework also aims to build the prototype in such a way that each security claim can be tested and supported by measured evidence.

39 The specific objectives of this project are as follows:

- ToapplyZeroTrust,request-levelaccesscontrolbycontinuouslyverifyingiden-tity, session, role, behaviour and risk for every file operation, including multi-factor authentication when the risk score is high.
- To inspect uploaded files with a multi-stage AI security engine that identifies mal-wareindicators,high-entropy(packedorencrypted)content,sensitivepersonal data and hidden payloads, and that maps findings to the MITRE ATT&CK frame-work.
- Toprotectfileconfidentialityusinghybridencryption,withAES-256-GCMfor file content and RSA-4096-OAEP for wrapping the per-file key, so that the server stores only ciphertext.
- To store encrypted content on IPFS through Pinata, with a local mock fallback, and to identify each stored file with a CID.
- To register file hashes, CIDs, signatures and permissions on a blockchain through Solidity smart contracts, giving an independent integrity reference.
- Tosupportcontrolledsharingwithrecipients,expiry,downloadlimitsandQR links, and to re-check the policy at every download.
- Tomaintainaprovenanceandaudithistoryforupload,share,revoke,download and denied-access events, and to provide a real-time security operations dashboard with automated compliance scoring.
- Toevaluatetheprototypeforfunctionalcorrectness,integrity,accesscontrol, threat-screening behaviour, retrieval time and usability.

1.3 Motivation for this Work

The motivation for this work arises from the gap between the number of security controls thatexistandthewaytheyareusuallycombinedineverydayfile-sharingtools.Afile

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passesthroughseveralrisksinitslife:theidentityandpermissionoftheusermustbe checked, the content may contain malware or sensitive information, the stored bytes must remainconfidential,alaterdownloadmustbecheckedforintegrity,andinvestigators needatraceableactivityrecord.Inatypicalcentralizedplatform,theseconcernsare handled by separate components, or are not handled at all.For example, an upload may be scanned by a separate process whose result is not preserved, storage may be readable by anyone who controls the server, and the audit trail may be an ordinary table that can be modified.When these controls do not share information with one another, an attacker or insider can exploit the weak link between them.

A second motivation is the growing demand for verifiable evidence. Regulations such as GDPR Article 32 call for security measures that are appropriate to the risk, including confidentiality, integrity, availability and resilience, and standards such as ISO/IEC 27001 and the NIST Cybersecurity Framework expect logging, cryptography and access control tobedemonstrable.Organizationsalsoneedtoreconstructincidents,whichrequires evidence of who performed an action, what happened, when it happened and which file and transaction were involved.At the same time, behavioural and geographic anomalies cannot be detected by static role-based access control alone.Decentralized storage and blockchain records offer a way to keep file content and integrity evidence outside a single administrator's control,while AI-assisted analysis offers a way to flag risk before a file isshared.Theseobservationsmotivatethestudyofanintegrateddesignthattreatsthe securitycontrolsaspartofthefilelifecycle,andthatisevaluatedthroughend-to-end tests rather than through component-level claims.

1.4 Methodology of the Project

ThemethodologyfordevelopingtheAegisCyberShieldfollowsastructured,layered and incremental approach.The work begins with requirement analysis,where the risks that a file faces during upload, storage, sharing and download are converted into concrete requirements:identityverification,permissionenforcement,contentinspection,confi -dentiality,integrityverificationandtraceability.Aliteraturesurveyofresearchpapers,

technicalstandardsandofficialsecurityguidanceiscarriedouttosupporteachdesign choice,andthesurveyisgroupedintothemessuchasZeroTrustandgovernance,de-

centralized storage,blockchain integrity,cryptography,AI threat detection,behavioural analyticsandsecurityoperations.Theoutcomeofthisstageisafive-layerarchitecture comprising presentation, gateway, intelligence, integrity and storage layers, together with defined trust boundaries that separate encrypted file bytes, integrity references and oper-ational metadata.

In the implementation stage, each layer is developed as an independent module.The presentationlayerisaReactandTypeScriptdashboard.ThegatewayisanExpress.js servicethatappliesasix-stageZeroTrustpipelinetoeachrequest.Theintelligence layerisaPythonFastAPIservicethatperformsmalware,entropy,personal-dataand steganographyscansandevaluatessessionrisk.TheintegritylayerconsistsofSolidity contracts, FileRegistry and DecentralizedID, executed on a local Hardhat EVM with an in-memory ledger fallback. The storage layer pins encrypted files to IPFS through Pinata, with a local mock fallback, while MongoDB (or a development mock) stores users, file records and activity records.During an upload, a file is scanned, encrypted with a fresh AES-256-GCM key, the key is wrapped with the RSA public key, the SHA-256 hash is computedandsignedwithECDSA,theciphertextispinnedtoIPFS,andthehashand CID are registered on the blockchain.

Thedownloadandsharingflowsapplythesamedisciplineinreverse.Whenauser requestsafile,thegatewayvalidatesidentityandpermission,checksownershiporthe active share policy, confirms access through the on-chain registry, fetches the ciphertext usingitsCID,verifiesthehash,unwrapstheAESkeyanddecryptsthefileonlyfor theauthorizedrecipient.SharingsupportsdirectusersharesandQRlinkswitha32-byterandomtoken,expiryandoptionaldownloadlimits,andthepolicyisre-checked on every download.Each operation generates a FileActivity record with the actor, time, outcome, device and location details, and a transaction reference where applicable.Real-timeeventsaredeliveredthroughWebSocket,andninesecuritytoolsareexposedto compatible AI clients through the Model Context Protocol.

Finally,theprojectfollowsavalidationanddocumentationapproach.Testcases aredefinedforauthentication,authorization,filevalidation,integrity,storageretrieval,

audit logging, performance and usability, and the input, expected outcome and observed resultofeverytestcaseisrecorded.Moduleprogressisestimatedprovisionallyandis to be validated against code, logs and demonstrable tests.The work is carried out over a

seventeen-week plan, in which Weeks 1 to 10 cover research, design and initial module development, and Weeks 11 to 17 cover integration, negative testing, evidence collection and documentation for the Phase 1 review.

Chapter 2

LITERATURE REVIEW

2.1 LITERATURE SURVEY

The literature survey covers 30 sources, comprising 16 research papers and reports and 14 standards and official documents.The sources are grouped into themes:Zero Trust and governance,decentralizedstorage,blockchainintegrity,cryptography,AIthreatdetection, user and entity behaviour analytics, security operations and forensics, and support-ing tools.The most relevant sources are summarized below, and the supporting standards are summarized in Section 2.2.

2.1.1 Zero Trust Architecture

- **Authors:**S. Rose, O. Borchert, S. Mitchell, S. Connelly
- **Source:**NIST Special Publication 800-207, 2020

This publication defines Zero Trust Architecture (ZTA) as a security approach in which no user, device or request is trusted solely because of its network location.Instead, every requesttoaccessaresourceisevaluatedusingtheidentityofthesubject,thestateof thedevice,thesensitivityoftheresourceandadefinedaccesspolicy.Thedocument describes logical components such as a policy engine, a policy administrator and a policy enforcement point, and discusses deployment variations and threats to the architecture.

Thecentralfindingisthataccessdecisionsmustbemadecontinuouslyandperre-quest,ratherthanonceatthenetworkperimeter.Perimeter-baseddefencesareweak againstadvancedpersistentthreatsandlateralmovement,becauseanattackerwhohas

entered the network inherits the trust that the perimeter grants.By moving enforcement close to each resource and re-evaluating context, a Zero Trust design limits what a single compromised account or host can reach.

In this project, the publication motivates the Express.js gateway, which acts as a pol-icy enforcement point.The gateway checks the token, source address, session state, role and attribute policy,behavioural score and risk for each request,and the same policy is re-checked on every download.

2.1.2 IPFS – Content Addressed, Versioned, P2P File System

- **Authors:** J. Benet
- **Source:** arXiv:1407.3561, 2014

This paper presents the InterPlanetary File System, a peer-to-peer distributed file system thatconnectscomputingdeviceswiththesamesystemoffiles.Filesarebrokeninto blocksthatareidentifiedbyacryptographichash,andtheblocksarelinkedtogether into a Merkle directed acyclic graph.The resulting Content Identifier (CID) depends on the content of the file and not on its location, which gives deduplication and makes any modification of the content detectable.

The paper also describes content-addressed retrieval through a distributed hash table andblockexchangebetweenpeers.AnimportantpracticalobservationisthataCID guarantees what the content is, but it does not by itself guarantee that the content remains available or that it is confidential. Availability depends on nodes that keep the data, which is why pinning services are used, and anyone who knows a CID can retrieve unencrypted content.

Forthisproject,IPFSisusedtoholdtheencryptedfileonly.Thefileisencrypted before it is pinned, the CID is recorded on the blockchain registry, and the pinning service (Pinata, with a local mock fallback) is used to keep the content available.

2.1.3 How to Time-Stamp a Digital Document

- **Authors:** S. Haber, W. S. Stornetta
- **Source:** Journal of Cryptology, 3(2), 1991

This early paper addresses the problem of proving that a digital document existed at a cer-tain time and has not been altered since.The authors propose time-stamping schemes in which the hash of a document is linked to previously issued time-stamps, so that creating a back-dated or altered record would require recomputing every later link.The scheme does not require the content of the document to be revealed to the time-stamping service, because only its hash is submitted.

The key idea is that a small, linked and publicly verifiable record can provide strong evidenceaboutamuchlargerdocument.Thisideaunderlieslaterhash-linkedledgers, including blockchains such as Bitcoin, which make past tampering detectable.

The project follows the same principle.The system anchors the SHA-256 hash and CID of each encrypted file on the blockchain registry and not the file itself.This limits cost, keeps the file content private and provides an independent reference against which a later download can be compared.

2.1.4 Blockchain-EnabledIntrusionDetectionandPreventionSystem of APTs within Zero Trust Architecture

- **Authors:**C. Alevizos et al.
- **Source:**IEEE Access, 10, 89270–89288, 2022

ThispaperproposesasystemthatcombinesaZeroTrustarchitecturewithblockchain technology to detect and prevent advanced persistent threat (APT) intrusions.The work
arguesthatAPTsaredifficulttodefendagainstwithperimetersecurityalone,andthat theshared,tamper-evidentstateofablockchaincansupporttrustworthydetectionand response across the components of a Zero Trust deployment.

ThepaperistheclosestpriorworktothepresentproJect,becauseitalsousesZero Trust and blockchain together for security.Its focus, however, is intrusion detection and prevention for APTs and not the complete life cycle of a shared file.The remaining gap, whichthepresentprojectaddresses,isafullfile-sharingworkflowinwhichscreening, encryption, content addressing, on-chain registration, access checks and provenance are bound together.

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2.1.5 Hyperledger Fabric:A Distributed Operating System for missioned Blockchains

- **Authors:**E. Androulaki et al.
- **Source:**EuroSys, 2018

ThispaperdescribesHyperledgerFabric,amodularblockchainplatformdesignedfor permissionedenterprisesettings.Unlikepublicblockchainsinwhichtransactionsare orderedandthenexecutedbyeverynode,Fabricusesanexecute-order-validatearchitecture.Chaincodeisfirstexecutedbyendorsingpeers,theresultingtransactionsare orderedbyanorderingservice,andeachpeerthenvalidatesthembeforecommitting them to the ledger.

The paper highlights that a permissioned ledger gives verifiable shared state and audit trailstoaknowngroupofparticipants,whileofferingbettercontrolofconfidentiality and performance than an open network.It also illustrates the design trade-offs between decentralization, throughput and privacy.

This work motivates the audit-trail design of the project and the decision to keep the on-chainrecordsminimal.Intheprototype,SolidityContractsrunonalocalHardhat EVM, but the same registry concept could later be moved to a permissioned ledger.

2.1.6 Applications of Blockchains in the Internet of Things:A Comprehensive Survey

- **Authors:**M. S. Ali et al.
- **Source:**IEEE Communications Surveys & Tutorials, 2019

This survey reviews how blockchain technology is applied in the Internet of Things and discussestheopenchallengesthataccompanytheseapplications,includingscalability, storage, privacy and transaction cost.The authors note that storing large volumes of data

directlyonablockchainisimpractical,becauseeveryfullnodemustreplicatethedata and the entries are difficult to remove once published.

Therelevanceofthesurveytothisprojectisthedesignimplicationthatonlycom-pact integrity references should be placed on the chain.The project therefore keeps file

contents encrypted on IPFS and registers only file hashes, CIDs, threat scores, signatures and permissions on the blockchain, which reduces cost and limits privacy exposure.

2.1.7 The Role of Machine Learning in Cybersecurity

- **Authors:** G. Apruzzese et al.
- **Source:** ACM Digital Threats:Research and Practice, 4(1), 2023

This article reviews how machine learning is used in cybersecurity and examines the practicallimitationsthatarisewhensuchsystemsaredeployed.Itdiscussesissuessuchas data quality, evaluation methodology, adversarial behaviour and the cost of false alarms, and it argues that machine-learning detectors should not be treated as infallible.

Theauthors'mainmessageisthatlearneddetectorsaremostusefulwhentheyare combinedwithothercontrolsandareevaluatedhonestly.Forthisproject,thefinding supports a hybrid design in which rule-based checks and analytics are combined, in which every AI finding is recorded with its evidence, and in which false positives and negatives are measured against test data rather than ignored.

2.1.8 A Mathematical Theory of Communication

- **Authors:** C. E. Shannon
- **Source:** Bell System Technical Journal, 27(3), 1948

Shannon'spaperfoundedinformationtheoryanddefinedentropyasameasureofthe information content, or unpredictability, of a source.Data whose bytes are close to uni-formly random has high entropy, whereas ordinary text and many structured formats have lower entropy.

This property is used in malware and ransomware analysis.Packed executables and encrypted content have unusually high byte entropy, so a high entropy score can indicate that a file is packed, encrypted or obfuscated.In the project, an entropy analyzer inside the AI security engine uses this idea to flag packed, ransomware-like files, and the result is treated as a risk indicator and not as a final verdict.

2.1.9 Toward Generating a New Intrusion Detection Dataset trusion Traffic Characterization

- **Authors:** I. Sharafaldin, A. H. Lashkari, A. A. Ghorbani
- **Source:** ICISSP, 2018

This paper introduces the CICIDS2017 dataset, which contains benign traffic and a range of common attacks generated in a controlled testbed, together with an analysis of traffic features and a characterization of the attacks.The dataset was created to overcome short-comings of older intrusion-detection datasets,such as outdated attack types and limited diversity.

In the project, the dataset is considered as a candidate benchmark for evaluating the detectionandbehaviour-analysis modules.The planned evaluationof the AI engine on benchmark data is part of the future work, because detection quality depends on data and must be demonstrated by measurement.

2.1.10 Keystroke Dynamics as a Biometric for Authentication

- **Authors:** F. Monrose, A. D. Rubin
- **Source:** Future Generation Computer Systems, 2000

This paper studies keystroke timing as a behavioural biometric.The way a person types, including the duration forwhich keys are held and the intervals between them,forms a patternthatcanbecomparedwithastoredprofiletosupportuserauthentication.The authorsevaluateclassificationmethodsontypingdataandshowthatsuchpatternscan provide useful identity evidence.

ArelatedlineofworkbyT.Bassproposesthatfusingdatafromseveralsensors improves intrusion detection.Together, these works motivate the behavioural part of the gateway.Intheproject,keystrokedynamicsandmousejitterarriveasHTTPheaders, and they are combined with device and geographic signals into one risk score between 0 and 100 per session.Multi-factor authentication is triggered when the score reaches 70 or more.

2.1.11 SecurityOperationsCenter:ASystematicStudyandOpen Challenges

- **Authors:** M. Vielberth et al.
- **Source:** IEEE Access, 8, 227756–227779, 2020

Thispaperpresentsasystematicstudyofsecurityoperationscentres(SOCs),covering theirpeople,processesandtechnology,anditidentifiesopenchallengessuchasalert overload,integrationoftoolsandlackofautomation.Thestudyshowsthateffective monitoring requires the integration of data from several sources into a coherent view.

This motivates the live dashboard, alerts and incident workflow of the project.File-sharing events, threat alerts and compliance information are presented to a SOC analyst in one place, and a set of cooperating agents (scanner, responder, forensics and compliance) automates routine analysis.

2.1.12 Getting Physical with the Digital Investigation Process

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- **Authors:** B. Carrier, E. H. Spafford
- **Source:** CERIAS Technical Report 2003-29, 2003

Thisreportproposesaprocessmodelfordigitalinvestigationthatisbasedontheinvestigationofphysical crimescenes.It frames digital activity as evidence that must be preserved, interpreted and documented through a defined process, so that conclusions can be defended.

For the project, this supports the design of the per-operation FileActivity record. Each action stores the event type, actor, IP address, geolocation, device fingerprint, recipient, expiry, transaction hash and outcome, so that a chain of custody can be reconstructed for any file.

2.2 Standards and Official Guidance Reviewed

Inadditiontotheresearchpapers,thefollowingstandardsandofficialdocumentswere reviewed.Each of them is mapped to a feature of the system.

- **NIST FIPS 197 (AES) and NIST SP 800-38D (GCM):** AES is a standard block cipher with 128-, 192- and 256-bit keys, and Galois/Counter Mode provides confi-dentiality together with an authentication tag.The system uses AES-256-GCM for each file, so that tampering with the ciphertext is detected on download.

- **RFC 8017 (PKCS #1 v2.2) and the Diffie–Hellman and RSA papers:** public-key cryptography lets parties share keys without a prior secret channel, and RSA-OAEP provides a secure padding scheme.The system uses RSA-4096-OAEP to wrap the per-file AES key for the recipient.

- **NIST FIPS 180-4 (SHA-2) and FIPS 186-5 (DSS):** SHA-256 digests are used to verify files against stored hashes, and ECDSA signs each file hash with the user's key.

- **NISTFIPS203(ML-KEM):**alattice-basedkey-encapsulationstandardderived fromCRYSTALS-Kyber. TheprojectsimulatesaKyber-1024KEMforpost-quantum readiness only; it is not deployed.

- **NISTCSF2.0,ISO/IEC27001:2022andGDPRArticle32:**thesedefineout-comesandcontrolsforriskmanagement,logging(control8.15),cryptography (control 8.24) and security of processing. The compliance scorecards of the system map its controls to these frameworks.

- **MITRE ATT&CK:** a catalogue of adversary tactics and techniques.Each alert of the AI engine is mapped to a tactic and technique identifier.

- **OWASP Top 10:2021:**identifies broken access control and cryptographic failures as leading web risks. The gateway therefore enforces server-side access checks and encryption by default.

- **Model Context Protocol specification and IBM Cost of a Data Breach Report 2023:**thefirstdefinesastandardwayforAIclientstocallexternaltoolsandis used to expose nine security tools, and the second provides the average breach cost figure of USD 4.45 million used for motivation.

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2.3 Research Gap

Thereviewedliteratureshowsthattheindividualbuildingblocksarewellestablished. Zero Trust, IPFS, blockchain, hybrid cryptography and machine-learning-based detection have each been studied in depth.The comparison in Table 2.1 is qualitative and is given only for discussion.It shows that existing approaches address only some of the required capabilities, and that prior work such as Alevizos et al. already combines Zero Trust and blockchainforintrusiondetection.Theremaininggapisintegrationatthelevelofthe complete file-sharing workflow.

Capability	Cloud sharing	IPFS only	Chain only	Aegis
Single point of failure	Yes	No	No	Partly
Content-addressedstor-age	No	Yes	No	Yes
Tamper-evident record	Varies	Partial	Yes	Yes
Encryptionbeforestor-age	Varies	No	No	Yes
Behaviourchecksatac-cess	Rare	No	No	Yes
Compliance monitoring	Varies	No	No	Yes

Table 2.1:Integration gap compared with existing approaches (qualitative)

ThegapaddressedbythisprojectisasingleworkflowthatbindsAIscreening,en-cryption,contentaddressing,on-chainregistration,accesschecksandprovenancefor afile-sharingservicethatisvisibletoasecurityoperationsteam.Theliteraturealso highlightspracticallimitationsthattheprojectmustaddress:aCIDdoesnotguarantee availability or confidentiality, on-chain data is difficult to remove, AI detection depends ondataqualityandproducesfalsepositivesandnegatives,andlogsalonedonotprove that every control is correctly enforced. These limitations are carried into the risk analysis and the testing plan.

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Chapter 3

SYSTEM DESIGN

Systemdesignistheprocessofdesigningtheelementsofasystemsuchasthearchi-tecture,modulesandcomponents,thedifferentinterfacesofthosecomponentsandthe data that flows through that system.The design of Aegis Cyber Shield separates user in-teraction, access enforcement, threat analysis, integrity records and file storage into five layers, so that each layer has a clear responsibility and a clear hand-off to the next.

3.1 System Architecture

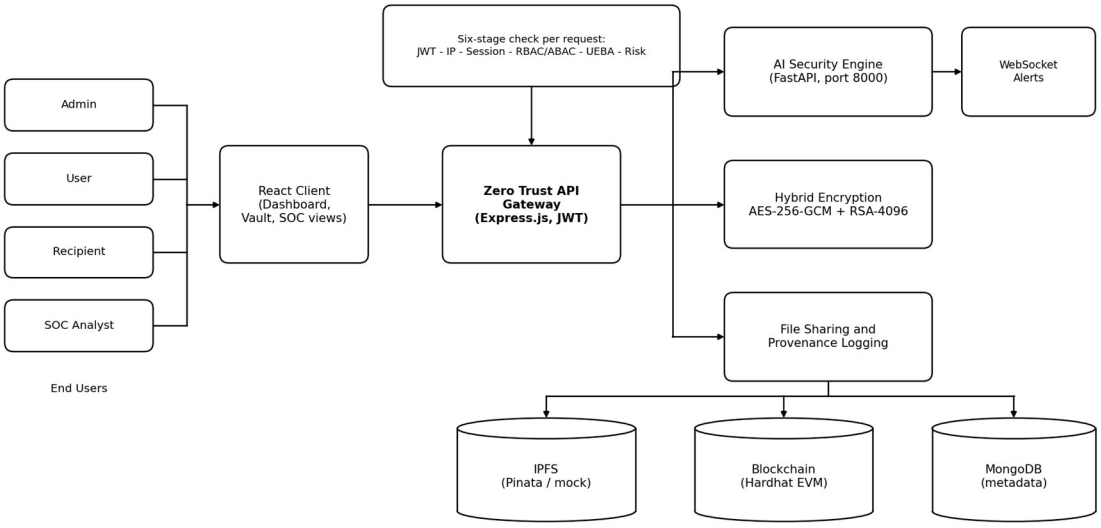


Figure 3.1:System Architecture of Aegis Cyber Shield

Thearchitecturehasfivelayers.ThePresentationlayerisaReactandTypeScript dashboard with more than fifteen pages (development port 3000).The Gateway layer is

anExpress.jsZeroTrustAPIgateway(port5000).TheIntelligencelayerisaFastAPI service for multi-stage scanning and behaviour analysis (port 8000).The Integrity layer is an EVM smart-contract registry with decentralized identity management (port 8545). TheStoragelayerisIPFScontent-addressedstoragethroughPinatawithamockfall-back.Supporting services include a MongoDB metadata store (port 27017) with a mock fallback, WebSocket event notifications, hash and audit records, and test scripts and logs. The ports quoted are development defaults.

Layer	Main responsibility	Output / hand-off
Presentation	Collectuseractionsand show status	Filerequest,scanresult and sharing state
Gateway	Authenticate, authorize and validate requests	Approvedrequestora recorded denial
Intelligence	Inspect file content and cal-culate risk indicators	Scan outcome and expla-nation
Integrity	Registerandverifyhash and CID references	Transactionorregistry result
Storage	Keep encrypted content ad-dressable	CID used for retrieval

Table 3.1:Responsibility of each layer

3.1.1 End Users and Actors

The system is designed around six actors, each with defined privileges.The **User** regis-

ters,logsin,uploads,downloadsandsharesfiles.The**Recipient**receivesfilesthatare shared with them through the shared inbox.The **Public** actor is an unauthenticated per-son who opens a QR share link, and can access only the single file that the link represents, subject to expiry and download limits.The **Admin** manages users,roles,IP allow-lists and system health.The **SOC analyst** monitors alerts, the threat map, forensic timelines and compliance scorecards. The **AI agent** accesses nine security tools through the Model Context Protocol.A role-based model, together with attribute-based rules, assigns each actoronlythefeaturesanddatathatarerelevanttotheirresponsibilities.Forexample, Auditor and Guest roles are limited to business hours.

3.1.2 Presentation Layer

ThePresentationlayeristheinterfacethroughwhichusersinteractwiththesystem.It is developed with React 18, TypeScript and Tailwind CSS, and offers pages for registra-tionandlogin,theencryptedfilevault,uploadwithscanresults,sharingandQRlinks, the shared inbox, the SOC dashboard, the threat map, analytics, file life cycle, forensics, compliancescorecardsandanAIcopilot.Toastnotificationsreplaceblockingalertdi-alogs,andthesharedinboxshowsActive,ExpiredandLimitedbadgessothattheuser canunderstandthesharingstateataglance.Thelayercollectsuseractionsandshows thestatusofscans,uploadsandshares,butitdoesnotmakesecuritydecisions;allen-forcement is performed on the server side.

3.1.3 Zero Trust Gateway

TheZeroTrustGatewayisthecoreaccess-controlcomponent.Everyrequesttoapro-tected endpoint passes through a pipeline of six checks, shown in Figure 3.2. JWT access tokens last two hours, and refresh tokens last seven days and rotate. Per-user IP allow-lists are supported, with a bypass for private addresses, and concurrent sessions are detected. The role-based and attribute-based policy engine limits Auditor and Guest roles to busi-ness hours.Keystroke dynamics and mouse jitter arrive as HTTP headers and are scored bythebehaviourevaluator,andanAIriskassessmentproducesacombined0to100 risk score per session.When the score reaches 70 or more, multi-factor authentication is triggered.A request that fails any stage is denied, and the denial is recorded in the audit log.

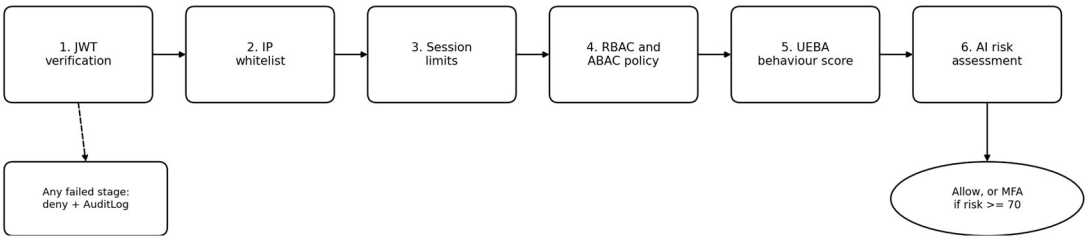


Figure 3.2:Zero Trust Gateway Pipeline

3.1.4 AI Security Engine

TheAISecurityEngineisaPython3.9+FastAPIservicethatinspectseachfilebefore itentersthenormalsharingpath.Itcombinesrule-basedheuristicswithanalytics,and everyfindingisreturnedwithanexplanation,sothatresultscanbetracedandfalse positivescanbestudied.Theenginecontainsfiveanalyzers.The**malwarescanner** applies heuristic detection.The **entropy analyzer** flags packed or ransomware-like files withunusuallyhighbyteentropy.The**personal-datadetector**searchesforsensitive identifiers such as social security numbers, card numbers, Aadhaar numbers and passport numbers.The**steganographyscanner**looksforhiddenarchivesinsideJPEGorPNG images.The **UEBA evaluator** calculates continuous session risk.Each alert is mapped toaMITREATT&CKtacticandtechnique.Theresponsetoaflaggedfile,whether rejection, quarantine or manual review, is a policy choice that is implemented and tested in the prototype.

3.1.5 Hybrid Encryption Module

The Hybrid Encryption Module protects file confidentiality and integrity.Encryption is carried out with the Node.js crypto library.A fresh AES-256-GCM key is generated for each file, and GCM supplies an authentication tag so that tampering with the ciphertext is detected on download.The AES key is then wrapped with the recipient's RSA-4096 public key using OAEP padding, so that a stored wrapped key cannot decrypt a file with-out the matching RSA private key.A SHA-256 hash of the file is computed and signed with ECDSA using the user's key.On the download path, the ciphertext is retrieved, the hashisverified,theAESkeyisunwrapped,andthefileisdecrypted.AKyber-1024 key-encapsulation mechanism is simulated to demonstrate post-quantum readiness, but it is not deployed.Key handling follows the principle that cryptographic strength depends onthegeneration,protection,storageandlifecycleofkeys,andnoprivatekeyorraw file content is placed in public blockchain records.

3.1.6 Blockchain Integrity Layer

The integrity layer provides an independent reference for the integrity of each file.Two Solidity contracts are developed with Hardhat and ethers.js. FileRegistry.sol pro-

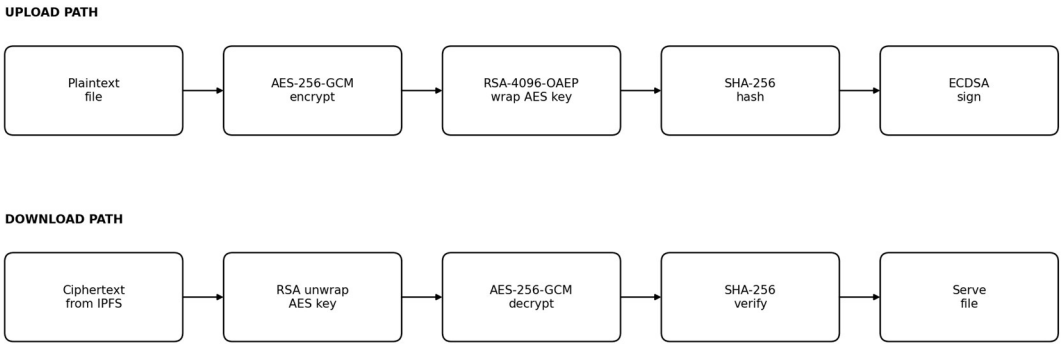


Figure 3.3:Hybrid Encryption:Upload and Download Paths

vides the functions registerFile,grantAccess,revokeAccess,verifyAccess and setEmer-gencyLock.DecentralizedID.solprovidesregisterDIDandgetDIDfordecen-tralizedidentities.ThecontractsexecuteonalocalHardhatEVM,andanin-memory ledger is used as a fallback when the EVM is not available.The chain stores file hashes,
CIDs,threatscores,signaturesandpermissions.FilecontentsstayencryptedonIPFS. Because entries on a blockchain are difficult to remove after publication, the records are kept minimal and do not contain personal information.

3.1.7 Decentralized Storage Layer

TheStoragelayerkeepstheencryptedfileaddressablethroughaCID.Encryptedfiles arepinnedtoIPFSusingthePinataservicesothattheyremainavailable,andalocal mock fallback allows development and demonstration when the service is not reachable. Because a CID guarantees content identity but not availability, retrieval from the pinned CID is part of the test plan.The ciphertext is the only data placed in this layer.

3.1.8 Database and Data Models

MongoDB (or a development mock) stores the operational metadata that is not suitable for the blockchain.Four main data models are used.The **User** record contains the user-name, bcrypt password hash, email, role, DID address, RSA public key, encrypted RSA privatekey,devicefingerprintbase,biometriccadence,whitelistedIPs,refreshtokens

andanactiveflag.The**FileDocument**containsthefilename,size,MIMEtype,CID, SHA-256 hash, owner, threat score, encrypted AES key, initialization vector, digital sig-

nature,blockchaintransactionhash,thesharedWithlist,andprovenanceandversion fields.The**FileActivity**recordstoresthefilereference,eventtype,actor,IPaddress, geolocation,successorfailurereason,transactionhashandmetadata.The**ThreatIntelligence** record stores the event type,severity,source and target address and location, threat score, confidence, MITRE tactic and technique, description, active state and tags. Each component stores only the information it needs.

3.1.9 File Sharing Module

Thesharingmodulesupportstwomechanisms.In*sharewithuser*,theownersendsa POST/sharerequestwiththetarget,durationandmaximumdownloads.Thegateway validates the owner, finds the recipient, adds a share entry with expiry and limits, logs the activity and notifies the recipient through WebSocket. In *QR link sharing*, the owner calls POST /generate-share-link,a 32-byte random token is created and stored in the shared-With list, and the link is rendered as a QR code.The recipient opens /share/:token, and the public-download endpoint checks expiry and limits before the file is served, without requiring a login.The policy is re-checked at every download, so that expired or revoked shares are denied.

3.1.10 Provenance and Audit Module

Every operation on a file produces a FileActivity record.The record contains the event type,actoridentity,IPaddressandgeolocation(city,country,latitude,longitudeand ISP),devicefingerprint,recipient,expiry,blockchaintransactionhashandwhetherthe operationsucceededorfailed.DownloadedPDFfilesarewatermarkedtosupportevidence preservation.These records support chain of custody and give investigators a clear upload,shareanddownloadtimelineforeachfile.Theauditlogcanadditionallybe anchored on the blockchain, so that it cannot be silently edited.

3.1.11 Agent Swarm, Threat Oracle and MCP Tools

Foursoftwareagentsruninparallel.The**scanner**agentlooksforthreats,brute-force attemptsandgeographicanomalies.The**responder**agenthandlesincidentresponse, includinganemergencylockdown.The**forensics**agentverifieschainofcustodyand

blockchain integrity.The **compliance** agent evaluates GDPR Article 32, ISO/IEC 27001 andNISTCSF.A**threatoracle**producesa72-hourforecastovereightthreatpatterns: ransomware,DDoS,phishing,exfiltration,bruteforce,zero-day,supplychainandinsider.In addition, nine security tools are exposed over JSON-RPC and server-sent events through the Model Context Protocol, so that compatible AI clients can query the system.

3.1.12 SOC Dashboard and Compliance Monitoring

TheSOCdashboardgivesasecurityanalystaliveviewofalerts,threatmaps,activityfeedsandscorecards,deliveredthroughWebSocketevents.Thecomplianceengine mapsimplementedcontrolstoGDPRArticle32,ISO/IEC27001:2022andNISTCSF 2.0outcomes,andproducesautomatedscores.Thesescoresdescribethebehaviourof the prototype against selected controls; they do not represent a formal certification.

3.2 Use Cases

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The use cases identified for the system are listed in Table 3.2.The complete use-case list is given in the project documentation.

Use case	Description	Main actor
UC1–UC2	Register and login	User
UC3	Upload and scan a file	User
UC4	Download and verify a file	User, Recipient
UC5–UC8	Sharewithauser,setexpiry andlimits,generateandscana QR link	User, Recipient, Public
UC9–UC10	SOC dashboard and threat map	SOC analyst
UC11–UC12	Agent swarm and threat oracle	SOC analyst
UC13–UC15	Analytics,filelifecycleand copilot	User, SOC analyst
UC16	Manage users and roles	Admin
UC17–UC18	Compliancereportandforensics	Admin,SOCana-lyst
UC19	AI agent analysis through MCP	AI agent

Table 3.2:Use-case summary (UC1 to UC19)

3.3 Security Controls and Compliance Mapping

Table 3.3 maps each implemented control to the standard that it supports.

Control	Implementation	Standard
Encryption at rest	AES-256-GCM per file	GDPR Art.32
Key management	RSA-4096-OAEP key wrapping	ISO/IEC 27001:2022 A.8.24
Identityverification	JWT and behavioural biometrics	NIST CSF 2.0 PR.AA
Access control	RBAC,ABACandZero Trust	NIST CSF 2.0 PR.AA
Audit logging	AuditLog with blockchain anchoring	ISO/IEC 27001:2022 A.8.15
Incident detection	AI scanning and UEBA	NIST CSF 2.0 DE.AE
Data integrity	SHA-256andon-chain hash	NIST CSF 2.0 PR.DS
Evidencepreservation	Watermarkingandcustody trail	NIST CSF 2.0 RS.AN

Table 3.3:Security controls and compliance mapping

3.4 API Surface

The gateway exposes REST endpoints grouped by area, as shown in Table 3.4.

Area	Endpoints
Authentication	POST/register,/login,/refresh,/logout,/wallet, /issue-vc, /verify-vc
Files	GET/list;POST/upload;GET/download/:id; POST /share; GET /shared-with-me; POST /generate-share-link; GET /public-share/:token;
Intelligence	GET/analytics/*,/threats/*,/forensics/timeline, /compliance/report; POST /copilot/query
Admin	GET/system-health,/users;POST/update-role, /update-ip
MCP	GET /sse; POST /messages

Table 3.4:API surface of the gateway

Chapter 4

METHODOLOGY

4.1 Requirement Analysis

The development begins with a requirement analysis phase in which the risks that a file faces during its life are converted into concrete requirements. Five risks are identified: the identity and permission of the user must be checked; the content may contain malware or sensitive information;the stored bytes must remain confidential;a later download must becheckedforintegrity;andinvestigatorsneedatraceableactivityrecord.Theserisks leadtosixrequirements,whicharesummarizedinTable4.1.Functionalrequirements includeregistrationandlogin,upload,download,sharing,revocation,monitoringand reporting.Non-functionalrequirementsincludesecurity,usability,performance,avail-ability of stored content and maintainability.

Thedesignprinciplethatfollowsfromthisanalysisisthatsecuritycontrolsshould connecttothefilelifecycleandshouldnotoperateasunrelatedfeatures.Afeasibility study confirms that the required technologies (React, Express.js, FastAPI, Solidity, Hard-hat, IPFS and MongoDB) are available as open-source or freely accessible services, and that stable local fallbacks exist for IPFS, the blockchain and the database for development and demonstration.

4.2 System Design

Theproposedsystemadoptsalayeredmodulararchitecture,describedinChapter3. Presentation,Gateway,Intelligence,IntegrityandStoragearedevelopedasseparate

Requirement	Expected system response
Identity verification	Authenticatetheuserbeforeprotectedfile operations.
Permission enforcement	Checkownership, recipientpermissions, link expiry and download limits.
Content inspection	Run configured file checks before the file enters the normal sharing path.
Confidentiality	Encrypt file bytes before external storage.
Integrity verification	Comparetheretrievedcontenthashwitha trusted recorded hash.
Traceability	Record upload, share, revoke, download and denied-access events.

Table 4.1:Requirements derived from the problem

modulesthatcommunicatethroughwell-definedinterfaces.Thedesignseparatesthree kindsofdataaccordingtotheirdifferentstorageneeds:encryptedfilebytesarestored onIPFSandidentifiedbyaCID,thefilehashandchosenmetadataarestoredinthe blockchainregistry,andusers,permissionsandactivityarestoredintheapplication database.Blockchainrecordsarekeptminimal,becauseentriesaredifficulttoremove after publication, and access permissions and encryption keys are never exposed through public metadata.Audit evidence refers to an event and its outcome and does not unnec-essarily duplicate private content.

4.3 Zero Trust Access Control

Accesscontrolisimplementedasasix-stagepipelineinthegateway.First,theJWT isverified.Second,thesourceaddressiscomparedwiththeper-userallow-list.Third, session limits and concurrent sessions are checked.Fourth, the role-based and attribute-basedpolicyisevaluated,includingbusiness-hourrestrictionsforAuditorandGuest roles.Fifth,thebehaviourscoreiscomputedfromkeystrokedynamics,mousejitter, device and geographic signals.Sixth, an AI risk assessment combines these signals into a score from 0 to 100, and multi-factor authentication is triggered at 70 or above.

The pipeline is applied to every request, including downloads, and not only to login.

ThisfollowstheZeroTrustprinciplethattrustisnevergrantedbecauseofaprevious successful request or network location.The outcome of each decision, whether allowed

or denied, is recorded in the audit log, so that the behaviour of the access-control layer can itself be inspected and tested.Because behavioural scoring can produce false positives, the design treats the score as an indicator that raises the level of verification, and not as an infallible decision.

4.4 AI-Based File Screening

File screening takes place in the upload path, before the file is encrypted and stored.The uploaded file is first validated for type, size and name in line with the OWASP guidance onfileuploads.Itisthenanalyzedbythemalwarescanner,theentropyanalyzer,the personal-data detector and the steganography scanner of the AI security engine.Each an-alyzer returns an indicator together with the evidence that produced it, and the indicators are combined into a threat score.Findings are mapped to MITRE ATT&CK tactics and techniques, which gives a consistent vocabulary for alerts.

The response to a flagged file is governed by policy.A file that passes the configured checks continues to normal encrypted storage, while a flagged file may be rejected, quar-antined or sent for manual review.The scan result and the actor and time information are storedforlaterreview.ThemethodologytreatsAIfindingsasriskindicatorsthatmust be tested with safe,controlled samples,and the limitations of detection,including false positives and false negatives, are reported openly.

4.5 Hybrid Encryption and Integrity Verification

Theencryptionmethodologyfollowsthehybridpatterninwhichasymmetriccipher protects the bulk data and an asymmetric cipher protects the symmetric key. On upload, a unique AES-256-GCM key and initialization vector are generated for the file, and the file is encrypted before it leaves the server.The AES key is wrapped with RSA-4096-OAEP, the SHA-256 hash is computed, and the hash is signed using ECDSA. The wrapped key, initialization vector and signature are stored in the file record, and the hash is registered on the blockchain.

Ondownload,thesystemfetchestheciphertextbyCID,recomputesandcompares theSHA-256hashwiththetrustedreference,unwrapstheAESkeyusingtherecipi-

Computer Science and Engineering Engg. 27 VAST, Thalakottukara

ent'sprivatekeyanddecryptsthefile.TheGCMauthentiationtagprovidesasecond, independentcheckthatthecipherthexthasnotbeenmodified.Ifeithercheckfails,the download is refused and the failure is logged.A stored wrapped key cannot decrypt a file without the recipient's RSA private key, which is not stored in a form the server can use.

4.6 Decentralized Storage and Blockchain Registration

After encryption, the ciphertext is pinned to IPFS through Pinata, and the returned CID is stored.The system then registers the file hash, CID, threat score and signature in the FileRegistrycontract,andthereturnedtransactionhashisstoredinthefilerecordand in the activity log.Permissions are granted and revoked through the same contract, and verifyAccess is called during download.An emergency lock can be set to freeze access to a file.Hardhat provides a local EVM for development and testing, and an in-memory ledger fallback keeps the workflow demonstrable when the EVM is not available.Smart-contract access conditions are tested locally, and no claim of a production audit is made.

4.7 System Integration and Automation

In this phase, the modules are integrated into one end-to-end workflow.The React client callstheExpressgatewayoverHTTPSandWebSocketSecure,thegatewaycallsthe FastAPIengineforscanningandriskassessment,theencryptionroutinespreparethe fileforIPFS,andethers.jsisusedtocallthesmartcontracts.Thegatewayalsowrites metadata and activity records to MongoDB. Real-time events are delivered to the dash-board through WebSocket, and the Model Context Protocol server exposes nine tools to AI clients.

Integrationcomplexityisoneofthemainrisksoftheproject,sosmallend-to-end testcasesandstablemockfallbacksareusedtoisolatefaults.Eachinterfaceistested firstwithsimpleinputs,andthecompletechainisthenverifiedbyanintegrationtest. Automation covers the whole path from upload to registration, and the agent

swarm runs periodic scanning, forensics and compliance checks without manual intervention.

4.8 Validation and Testing

The validation plan defines test cases for each security claim, and for each test the input, expected outcome and observed result are recorded.Authentication is tested with valid and invalid login and token scenarios.Authorization is tested by confirming that a non-ownerisdeniedandthatanexpiredorrevokedshareisdenied.Filevalidationuses an unsupported type, an oversized file and a suspicious but safe test sample.Integrity is tested by confirming that the recomputed hash of an unchanged file matches the registered reference, and that a modified file does not.Storage is tested by retrieving the CID while thecontentispinned.Theaudittrailistestedbyperformingupload,share,revoke, download and failed-access actions and confirming that each event is recorded.Usability is checked by asking whether a user can understand the scan outcome and sharing status.

Table 4.2 lists what will be measured.Safe, controlled test files and labelled samples are used for threat screening, and no harmful content is used.

Measure	How to test	Evidence to record
Upload cor-rectness	Uploadallowedsample files	Successorfailure,file size and timestamp
Integrity	Compare original and re-trieved hashes	Expected and actual hash match
Access control	Tryowner, authorized recipientandunautho-rized user	Allow or deny outcome for each case
Threatscreen-ing	Usesafe,controlledtest files or labelled samples	Detectionoutcomeand reason
Retrieval time	Repeat download for files of known sizes	Filesizeandelapsed time
Audit trail	Performupload,share, revoke and download ac-tions	Event record, actor time and outcome

Table 4.2:Evaluation measures

4.9 Performance Evaluation and Optimization

Performanceevaluationrecordstheuploadanddownloaddurationagainstthesizeof each test file, as well as the time taken by individual stages such as scanning, encryption,

IPFSpinningandblockchainregistration.Thesemeasurementsidentifywhichstage dominatesthedelayandwhetheritisacceptableforinteractiveuse.Retrievaltimeis measured by repeated downloads of files of known size.

Possibleoptimizationsincludestreamingencryptionforlargefiles,asynchronous registrationofhashessothattheuserisnotkeptwaitingforatransaction,cachingof access-checkresultsforshortperiods,anddatabaseindexingforactivityqueries.The performanceofalocaldevelopmentnetworkdoesnotrepresenttheperformanceofa public blockchain, and this limitation is stated when results are reported.

4.10 Project Schedule

The work is organized as a weekly plan. Weeks 1 to 10 were recorded in the project diary, and Weeks 11 to 17 form the work plan up to the Phase 1 review on Friday, 23 October 2026.The plan is summarized in Table 4.3.

Week	Dates	Activity
1	1–5 Jul	Researching the topic
2	6–12 Jul	Idea confirmation
3	13–19 Jul	Requirement analysis and Zero Trust
4	20–26 Jul	Five-layer architecture and data-flow design
5	27 Jul – 2 Aug	Dashboard, API planning and database setup
6	3–9 Aug	AI engine integration and risk assessment
7	10–16 Aug	Blockchain integrity and IPFS storage design
8	17–23 Aug	System module integration and provenance
9	24–30 Aug	Module-level testing and security compliance
10	31 Aug – 6 Sep	Overall review and documentation
11	7–13 Sep	Continuemoduleintegrationandresolveinter-face issues
12	14–20 Sep	Validate the upload, storage and hash-registration path
13	21–27 Sep	Checkaccesscontrol,sharingpermissionsand expiry rules
14	28 Sep – 4 Oct	Run negative tests and review error handling
15	5–11 Oct	Preparearepeatabledemonstrationandcollect test evidence
16	12–18 Oct	Consolidatereport,diagramsandpresentation material
17	19–23 Oct	Confirmimplementedmodulesandpreparefor the Phase 1 review

Table 4.3:Weekly project schedule (Weeks 1 to 17)

Chapter 5

DATA FLOW

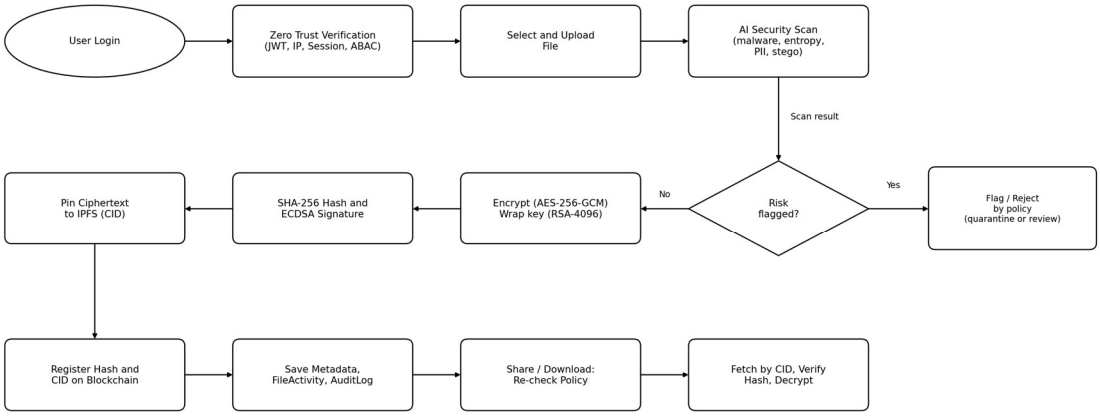


Figure 5.1:Data Flow Diagram

5.1 Description

Theflowchartillustratesthecompleteworkflowofthesystem,describinghowauser request is transformed into a protected, registered and verifiable file.The process begins withuserlogin,wheretheuseraccessesthesystemusingsecurecredentials.Thisis followed by Zero Trust verification, in which the JWT, source address, session, role and attribute policy and behavioural risk are evaluated for the request.This initial stage plays a crucial role in maintaining security, privacy and controlled interaction.

After successful verification, the user selects and uploads a file. The file is passed to the AI security engine, which performs malware, entropy, personal-data and steganogra-

phy scans and returns a threat score with an explanation. If the configured policy flags the file, the file is rejected, quarantined or sent for manual review, and the event is recorded. Only a file that passes the policy continues along the normal storage path,as shown by the decision step in the diagram.

For an accepted file, the system generates a unique AES-256-GCM key, encrypts the file, and wraps the key with RSA-4096-OAEP. The SHA-256 hash of the file is computed and signed using ECDSA. The ciphertext is then pinned to IPFS, which returns the CID, and the hash and CID are registered on the blockchain registry.Finally, the file metadata, theFileActivityrecordandtheAuditLogentryaresavedinthedatabase,andtheuser receives a confirmation.

When the file is later shared or downloaded, the policy is re-checked on every request. The system validates ownership or the active share, confirms access through the registry, fetchestheciphertextbyitsCID,verifiesthehash,anddecryptsthefileonlyforthe authorized recipient.Success or failure is recorded in the activity log in both cases.

Overall,thisworkflowrepresentsalayered,verifiableapproachtofilesharing.Encryptionisperformedbeforeexternalstorage,filecontentsareneverplacedonthe blockchain, and enough identifiers are recorded to compare a fetched file with its trusted hash.

5.2 File Upload Sequence

The upload sequence is shown in Figure 5.2.The user sends a POST request to the gate-way, which performs the Zero Trust checks and calls the AI engine. After the scan returns a threat score, the gateway encrypts, wraps, hashes and signs the file, pins the ciphertext to IPFS, registers the hash and CID on the blockchain, and saves the FileDocument, FileActivityandAuditLogrecordsinMongoDBbeforereturninga201Createdresponse. The scan result and the actor and time information are stored for later review.

5.3 File Download Sequence

The download sequence is shown in Figure 5.3. The user requests a file with GET /down-load/:id.The gateway applies the Zero Trust checks and verifies ownership or the active

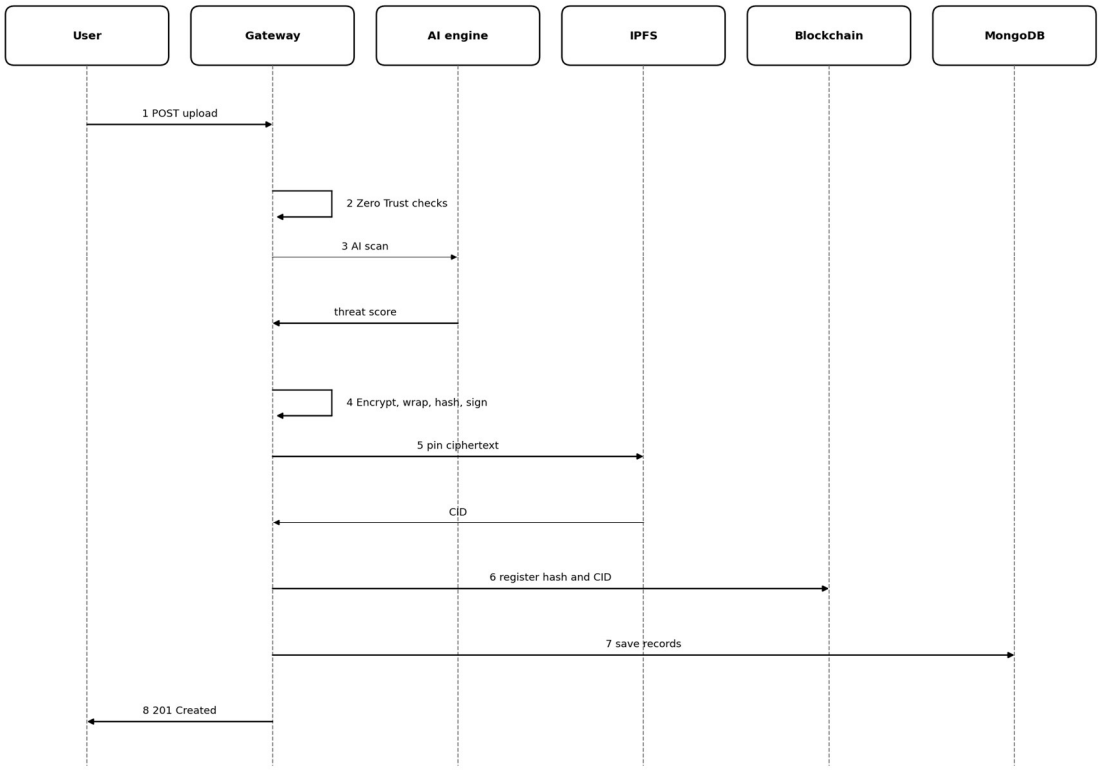


Figure 5.2:File Upload Sequence

share, and calls verifyAccess() on the blockchain registry.The ciphertext is fetched from IPFS by its CID, and the gateway checks the SHA-256 hash, unwraps the AES key and decrypts the file.For PDF files, a watermark is added.The gateway then updates down-load statistics and the AuditLog, and returns the file to the authorized user.

5.4 File Sharing Flow

Sharing follows two paths.In the first, the owner selects a recipient, and the gateway val-idates the owner's rights and the selected policy, sets the expiry and an optional download limit, adds the share entry, logs the grant, and notifies the recipient through WebSocket. Inthesecond,theownergeneratesaQRlink:a32-byterandomtokeniscreatedand stored,andthelinkisdisplayedasaQRcode.Apersonwhoopensthelinkcanreach onlythatfile,andthepublic-downloadendpointcheckstheexpiryanddownloadlimit before serving it.In both cases the policy is re-checked on every download, and grants, revocations and access attempts are logged.

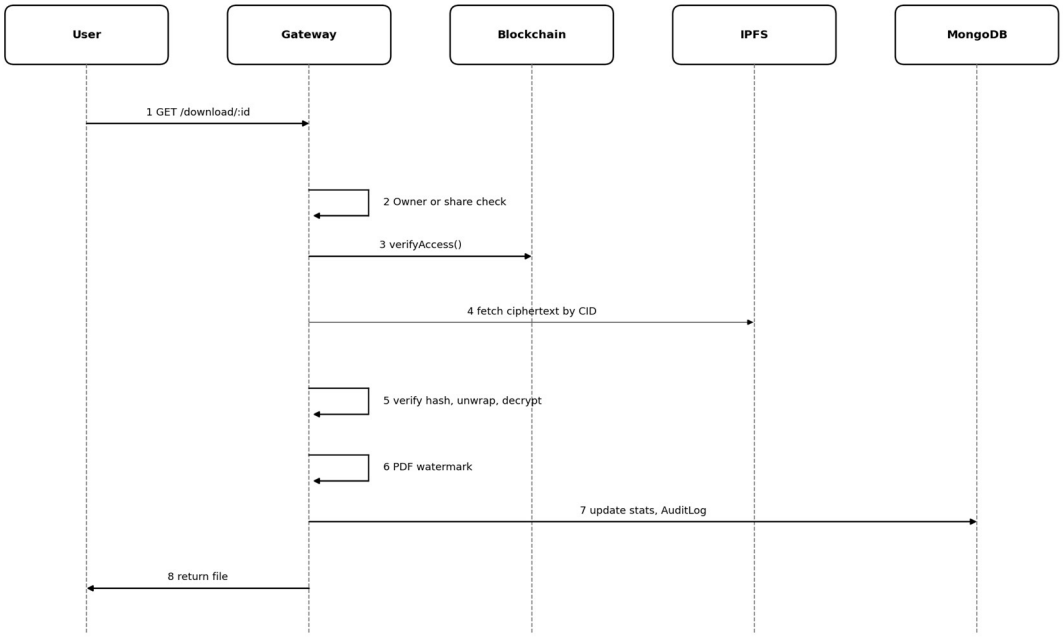


Figure 5.3:File Download Sequence

Chapter 6

WORK FLOW

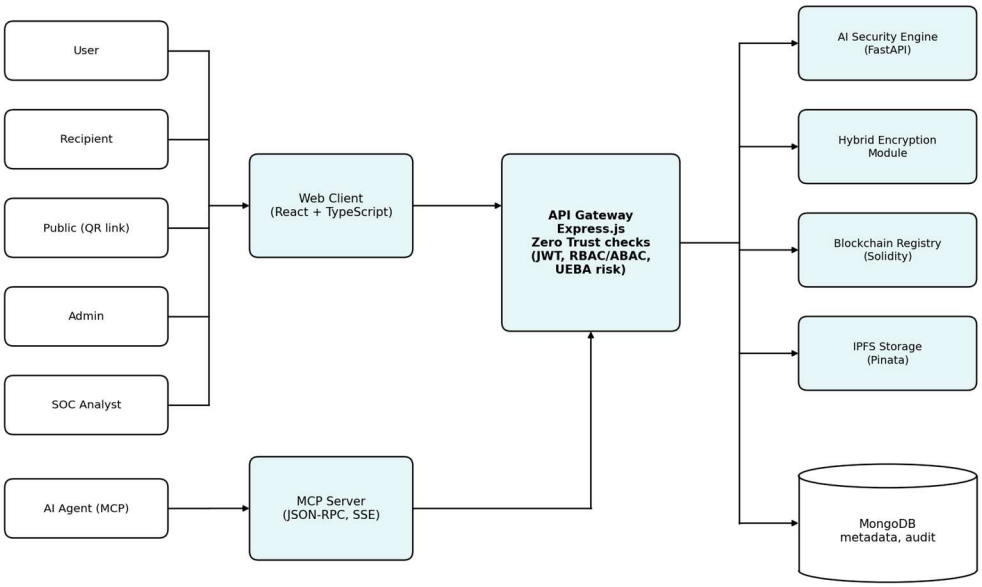


Figure 6.1:Work Flow Diagram

6.1 Description

The workflow diagram illustrates a layered file-sharing system that handles interactions from different types of actors and converts them into protected, registered and traceable files.On the left side, the diagram shows the actors:User, Recipient, Public (through a QR link), Admin, SOC Analyst and an AI agent.Each actor has a distinct responsibility. Usersupload,downloadandsharefiles,recipientsreceivesharedfiles,apublicvisitor can open only a single link, admins manage users and roles, SOC analysts monitor alerts

and compliance, and AI agents analyze the system through the Model Context Protocol.

Noneoftheseactorsinteractdirectlywiththeinternalservices;instead,theyoperate through client interfaces.

Human actors use the Web Client, which is the React and TypeScript front end.The clientcollectsuserinputssuchasfiles,sharesettingsandapprovalsandsendsthemas requests to the API Gateway.AI agents reach the system through the MCP server, which communicatesbyJSON-RPCandserver-senteventsandpassesrequeststothesame gateway.TheAPIGatewayisasingleentrypointintothebackend.Ithandlesrequest routing, authentication, validation, Zero Trust checks and risk assessment, and it ensures that requests are processed securely without exposing internal services directly to users.

Once the gateway approves a request, it forwards the request to the appropriate back-end modules.The AI Security Engine inspects file content and calculates risk indicators. The Hybrid Encryption Module encrypts the file, wraps the key, computes the hash and signsit.TheIPFSStoragemodulepinstheencryptedcontentandreturnsaCID,and the Blockchain Registry records the file hash, CID and permissions so that integrity can be verified later.All operational data, including user information, file metadata, activity records and audit logs, is stored in the Database.

Theoverallflowisthereforecleanandlayered:actorsinteractthroughclients,re-quests are managed centrally by the gateway, processing is handled by specialized mod-ules,andresultsarestoredanddeliveredwithanaudittrail.Thismodulararchitecture improves maintainability, allows independent testing of each component and supports fu-ture improvements, such as running the registry on a permissioned ledger, using several pinning providers, or adding evaluated machine-learning models to the AI engine.

Chapter 7

SYSTEM REQUIREMENTS

7.1 Hardware Requirements

7.1.1 Processor

The Aegis Cyber Shield prototype runs several services at the same time,including the Reactdevelopmentserver,theExpress.jsgateway,theFastAPIengine,alocalHardhat blockchain node and a MongoDB instance.In addition, it performs cryptographic oper-ationsandfilescanning.Amulti-coreprocessoristhereforeneeded.Aminimumcon-figuration of an Intel Core i5 or an equivalent AMD Ryzen 5 processor is recommended, as it provides adequate performance for development, testing and a demonstration with a small number of users.

Forbetterperformance,particularlywhenscanninglargefiles,runningautomated testsorcompilingsmartcontracts,ahigher-endprocessorsuchasanIntelCorei7or AMD Ryzen 7 is preferred.More cores and higher clock speeds shorten the time taken byAESandRSAoperations,entropyanalysisandcontracttests,andmulti-threading helps the system manage several requests without a noticeable loss of responsiveness.

7.1.2 Random Access Memory (RAM)

A minimum of 8 GB RAM is needed to run the full development stack.Memory is used bytheNode.jsandPythonruntimes,thelocalHardhatEVM,MongoDB,thebrowser-

based dashboard and the temporary buffers that hold a file during scanning and encryp-tion. If less memory is available, the services may need to be run with the mock fallbacks.

Forimprovedperformance,16GBRAMisrecommended.Theadditionalmemory allows larger files to be scanned and encrypted in memory, reduces swapping when sev-eral services and tools run together, and gives the system headroom for the AI engine and for concurrent sessions during demonstrations.

7.1.3 Storage

A minimum of 250 GB of available storage is recommended for the source code, depen-dencies,containerorvirtual-environmentfiles,thelocalblockchaindata,thedatabase, testfilesandlogs.Solid-StateDrives(SSD)arepreferredoverHardDiskDrivesbe-causetheyprovidefasterreadandwritespeeds,whichspeedsupfilehandling,builds and database access.

Encrypted files are stored on IPFS, and the local node or the mock fallback requires additional space proportional to the number and size of test files. For long-term use, a 512 GB SSD or larger is recommended, together with regular backups of the database and of the keys required for decryption.Losing the keys means losing access to the encrypted files, so key backup is a real operational need.

7.1.4 Networking Requirements

The prototype is a web-based system and needs a stable internet connection for the Pinata pinning service, for downloading dependencies and for access by remote clients.A mini-mum speed of 5 Mbps is required for standard use, and 10 Mbps or more is recommended when several users upload and download files at the same time, because the pinning and retrieval time depends on network speed. Clients must connect through HTTPS and Web-Socket Secure so that data and alerts are protected in transit.

Latencyandreliabilitydirectlyaffecttheuserexperienceduringuploads,because several steps, such as scanning, pinning and registration, are executed in sequence. When the network is not available, the local mock fallback for IPFS allows the workflow to be demonstrated, but it does not demonstrate decentralized availability.

7.1.5 Client-Side Requirements

Endusersaccessthesystemthroughawebbrowserandnospecializedsoftwareisrequired.Eachclientshouldhaveatleastadual-coreprocessorand4GBRAM,anda modern, up-to-date browser such as Google Chrome, Mozilla Firefox or Microsoft Edge isrequiredtosupportthedashboard,WebSocketconnectionsandtheWebCryptofeaturesusedbytheinterface.AFullHDdisplay(1920×1080)isrecommendedforthe SOCdashboard,chartsandthreatmap,andastableinternetconnectionisneededfor real-time alerts.

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7.2 Software Requirements

7.2.1 Operating System

The system is designed to be cross-platform.Development and demonstration are possi-ble on Windows 10/11 and on Linux distributions such as Ubuntu 22.04 LTS. A Linux-basedserverispreferredfordeploymentbecauseofitsstability,securityandefficient handling of web server and database workloads, while Windows is convenient for devel-opment, testing and debugging.

7.2.2 Frontend Technologies

ThefrontendisdevelopedwithReact18andTypeScript,whichprovidereusable, stronglytypedcomponentsandefficientrendering.TailwindCSSisusedforstyling andresponsivedesign,andthedashboardpresentscharts,athreatmapandlivealert feeds.CommunicationwiththegatewayusesHTTPSforrequestsandWebSocketSecureforreal-timeevents.Toastnotificationsareusedforuserfeedback,andQRcodes are generated for share links.

7.2.3 Backend Technologies

ThegatewayisimplementedwithNode.jsandExpress.jsinTypeScript.JSONWeb Tokensareusedforauthentication,Multer handles fileuploads,and the Node.js

crypto libraryperformsAES-256-GCM,RSA-4096-OAEP,SHA-256andECDSAoperations.

ThewslibraryprovidesWebSocketsupport,andtheModelContextProtocolSDKexposes nine tools to AI clients.A RESTful API structure keeps the frontend and backend modular and independent.

7.2.4 AI Engine

The AI security engine is written in Python 3.9 or later using FastAPI. It uses scikit-learn together with rule-based analyzers for malware heuristics, entropy analysis, personal-data detection,steganographydetectionandbehaviourscoring.Theengineisdeployedasa separate service (development port 8000) so that it can be scaled and tested independently of the gateway.

7.2.5 Blockchain and Storage

Smart contracts are written in Solidity and are compiled, tested and deployed with Hard-hat.The ethers.js library connects the gateway to the contracts, and a local Hardhat EVM (port 8545) is used for development, with an in-memory ledger as a fallback. File storage usesIPFSthroughthePinatapinningservice,withalocalmockfallback.Metadatais storedinMongoDB(port27017),withadevelopmentmockavailablewhen adatabase server is not installed.

7.2.6 Development and Deployment Tools

Development uses Visual Studio Code as the editor, Git and GitHub for version control,
PostmanfortestingAPIendpoints,andHardhat'stestframeworkforsmartcontracts.
Testscriptsandlogsrecordtheresultsofeachtestcase.Fordeployment,theservices canberunonalocalserverfordemonstration,oroncloudplatformssuchasAWS, MicrosoftAzureorGoogleCloud.ContainerizationwithDocker,andareverseproxy such as Nginx for TLS termination, can be used to give consistent environments.Secrets suchaskeysandtokensmustnotbehard-codedandaresuppliedthroughenvironment configuration.

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Chapter 8

EXPECTED RESULT

8.1 Expected result

Theproposedsystemisexpectedtogiveusersafile-sharingworkflowinwhichevery file is verified, screened, encrypted, stored and registered before it is shared.An upload should produce a CID and a blockchain transaction reference, and the file hash should be availableforlatercomparison.Adownloadbyanauthorizedusershouldsucceedonly after the policy check and hash verification, and a download by a non-owner, or through anexpiredorrevokedshare,shouldbedeniedandrecorded.Aflaggedfileshouldbe handled according to the configured policy, and the scan result should be preserved with its explanation.The audit trail should show a clear upload, share and download timeline, andtheSOCdashboardshouldpresentalertsandcompliancescores.Theresultsof theprototypewillbereportedthroughtestevidenceandarenotintendedtoestablish enterprise-scale security or compliance.

8.2 Implementation Progress

ImplementationprogressatthetimeofPhase1issummarizedinTable8.1.Theper-centages are provisional estimates for the eight modules and the overall average is about 70%.They are to be validated against the code, screenshots, logs and demonstrable tests before the review, and should not be read as measured results.

Module	Components	Progress
Zero Trust gate-way	JWT,IP,session,RBAC/ABAC, UEBA	75%
Hybrid encryption	AES-256-GCM, RSA-4096, SHA-256, ECDSA	70%
AI security engine	Malware, entropy, PII,stego, UEBA	70%
Smart contracts	FileRegistry.sol, Decentralize-dID.sol	65%
IPFS storage	Pinatapinning,localmockfall-back	65%
SOC dashboard	Live alerts, charts, scorecards	70%
MCP tools	NinetoolsoverJSON-RPCand SSE	65%
Provenance	FileActivity, GDPR,ISOand NIST scoring	75%
Overall	Average of the eight modules	**about 70%**

Table 8.1:Module progress (provisional estimates)

8.3 System Implementation Results

8.3.1 Web Interface Implementation

Thefollowingscreensillustratethewebinterfaceandthemainfunctionalitiesofthe implemented prototype.The images shown here are placeholders and are to be replaced by the actual screenshots of the running system.

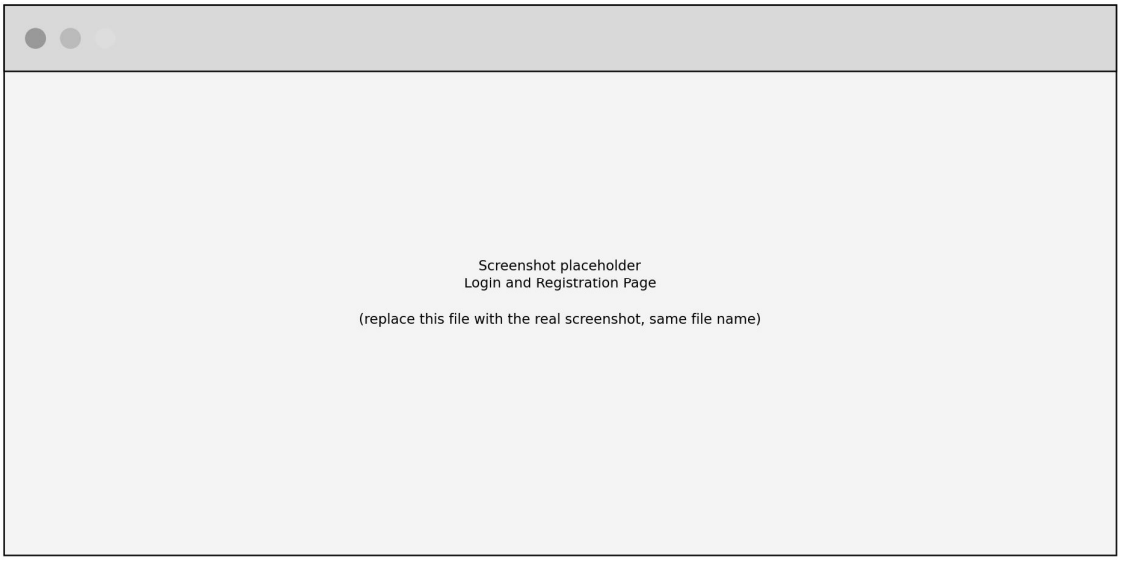


Figure 8.1:Login and Registration Page

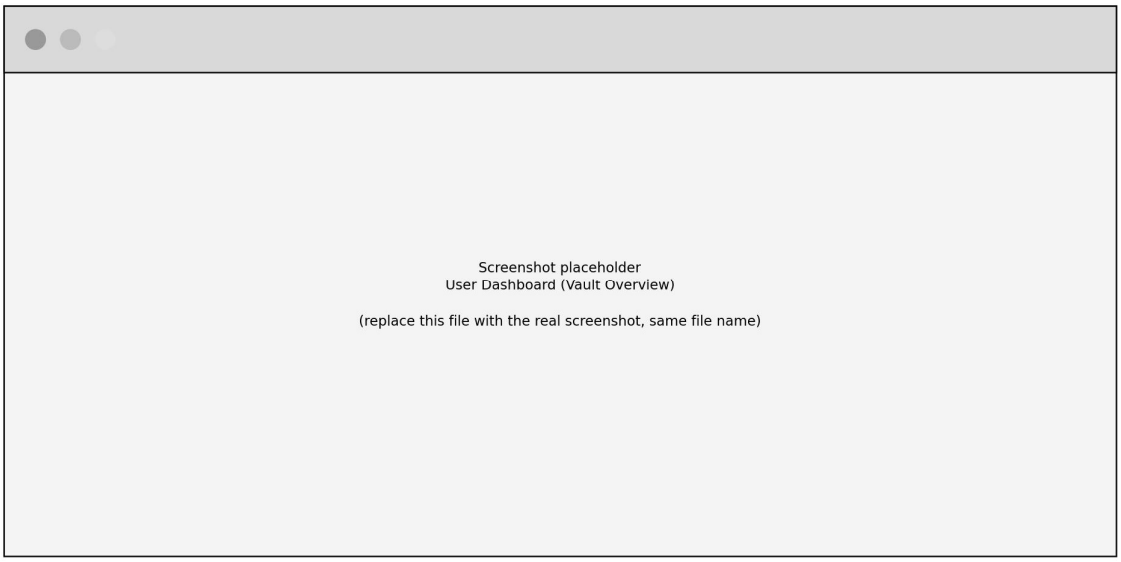


Figure 8.2:User Dashboard

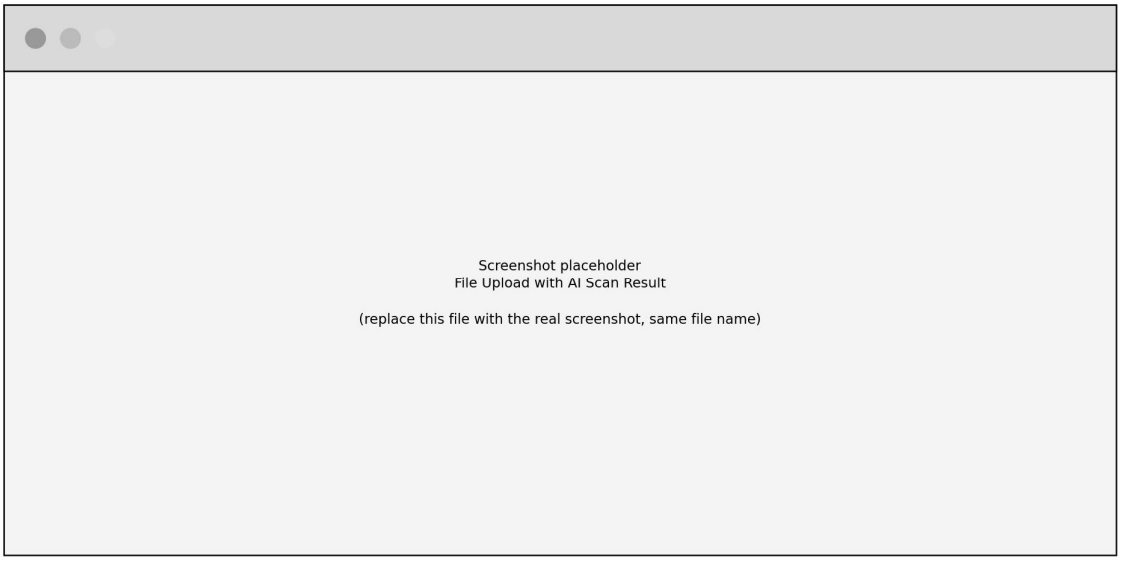


Figure 8.3:File Upload with AI Scan Result

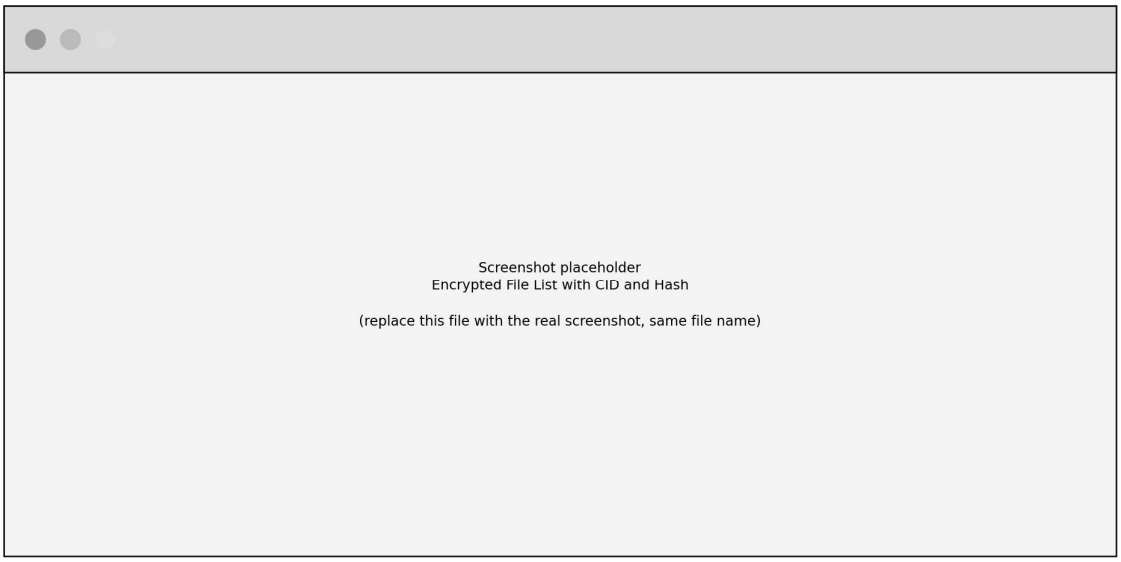


Figure 8.4:Encrypted File List with CID and Hash



Figure 8.5:Share Dialog with Expiry and QR Link



Figure 8.6:Shared Inbox

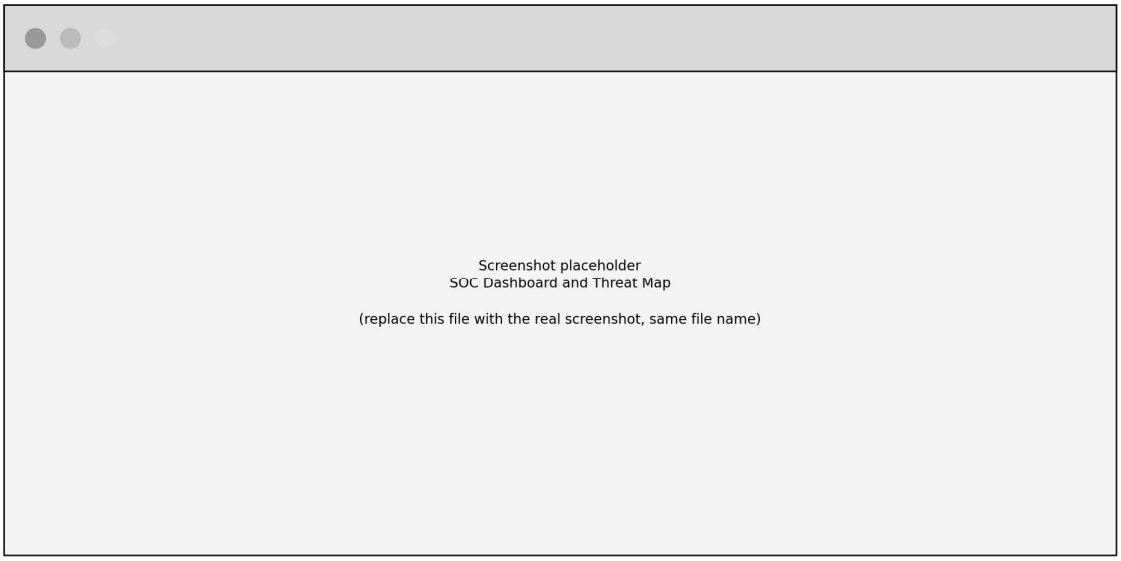


Figure 8.7:SOC Dashboard and Threat Map

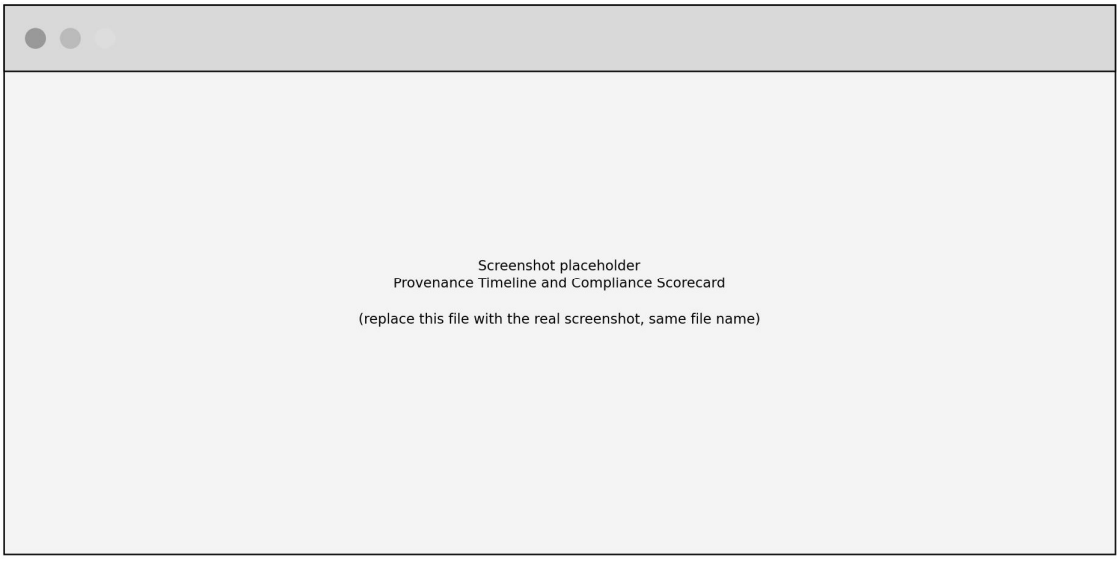


Figure 8.8:Provenance Timeline and Compliance Scorecard

8.4 Risks and Limitations

The main risks identified for the project and their mitigations are as follows.

•**IPFSavailability:**contentcanbeunavailableifnonodeprovidesit.Pinningis

used and retrieval is tested.

•**AI false positives and negatives:**transparent rule-based indicators are used,and

detection limitations are reported.

•**Key loss or exposure:**access to keys is restricted, secrets are not hard-coded and

the key life cycle is documented.

•**Blockchain privacy:**personal information and raw file content are kept off-chain.

•**Smart-contractdefects:**accessconditionsaretestedlocally,andnoproduction

audit or certification is implied.

•**Integration complexity:** small end-to-end test cases and stable mock fallbacks are

used where needed.

•**Prototype scope:** a working academic demonstration does not establish enterprise-

scale security or compliance.

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Chapter 9

CONCLUSION AND FUTURE WORK

9.1 Conclusion

File sharing is a high-value target, and centralized platforms depend heavily on a single server and on the practices of administrators.Aegis Cyber Shield addresses this problem by combining Zero Trust access control, AI-assisted threat screening, hybrid encryption, IPFSstorageandblockchainintegrityreferencesinonefile-sharingworkflow.Thelit-erature survey of 30 sources supports each individual technology and, at the same time, highlights practical trade-offs in detection quality, key management, availability and au-ditability.

Themaincontributionoftheprojectisanintegratedandtestableworkflow,notthe invention of the underlying technologies.Phase 1 has established the requirements,the five-layerarchitecture,theupload,downloadandsharingflows,thedatamodelsand theinitialmoduleimplementation.Thedesignkeepsfilecontentsencryptedandoff-chain, records minimal integrity references on the blockchain, re-checks policy at every download, and produces a provenance record for each operation.

Theworkisnotcomplete.Themoduleprogressisaprovisionalestimateofabout 70%,whichistobevalidatedbycode,logsandtests.Thenextstepsaretocomplete end-to-end integration, to test negative paths, to measure results, and to update the doc-umentationfromevidence.Decentralizationalonedoesnotguaranteeconfidentiality, availabilityorcorrectauthorization,andthesecontrolsmuststillbeimplementedcare-fully and tested.

9.2 Future Scope of Work

The system can be further improved and expanded in the following ways:

- **Evaluation of the AI Engine on Benchmark Data**
  - –Evaluate the detection modules on labelled datasets such as CICIDS2017 and on safe, labelled file samples.
  - –Measure precision, recall and the false-positive rate, and tune the risk thresholds.
  - –Add evaluated machine-learning models alongside the rule-based checks.
- **Stronger Key Management**
  - –Move key storage to a hardware security module or a managed key service.
  - –Add key rotation, recovery and revocation procedures.
  - –Document the complete key life cycle in line with NIST SP 800-57.
- **Post-Quantum Readiness**
  - –Replace the simulated Kyber-1024 keyencapsulation with a deployed ML-KEM implementation based on NIST FIPS 203.
  - –Evaluate hybrid classical and post-quantum key wrapping.
- **Production-Grade Blockchain and Storage**
  - –Deploy the registry on a test network or a permissioned ledger such as Hyperledger Fabric.
  - –Use multiple pinning providers or a private IPFS cluster to improve availability.
  - –Perform an independent audit of the smart contracts.
- **Scalability and Performance Optimization**
  - –Support streaming encryption and chunked uploads for very large files.
  - –Register hashes asynchronously and batch blockchain transactions.

–Carry out load testing with many concurrent users.

- **Usability and Platform Extension**

  –Develop mobile applications for sharing and for receiving alerts.

  –Improvetheexplanationofscanoutcomesandsharingstatusfornon-technical users.

  –Integrate with existing identity providers and organizational directories.

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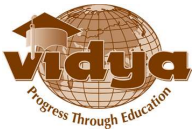
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